Disease Detection Based on Nail Color Analysis Using Image Processing

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Abstract—- There have been countless advancements in medical research, and a wide variety of diagnostic procedures for human body problems have been developed. Pathological testing, on the other hand, is time-consuming, painful, and difficult to employ in illness detection. Currently, the patient's nail is scanned and the image is divided. Experiment results demonstrate a considerable impact. The major drawback of this technique is that variations in blood flow may be detected by looking at the color of the image rather than segmentation. Dermatology studies have shown that nail color changes are a good indicator of a variety of health issues, and our suggested method is to use nail color analysis to look for signs of illness. An abnormality in one's health is indicated by a change in the color of one's nails. Heavy equipment and scanning processes are not necessary to take pictures of human nails for use in detection. In contrast to human sight, computer image processing can distinguish even the tiniest differences in color and hence provides more exact and accurate findings. Machine learning algorithms such as the CNN (Convolution Neural Network) algorithm are used in the suggested system, which takes an image and assigns value to distinct items in the picture. Using Picture Processing, an image is digitally transformed and a few procedures are carried out to extract valuable information. To identify nail traits, image analysis and digital image processing techniques such picture acquisition, image pre-processing, segmentation, and feature extraction are utilized.. Based on variations in nail color, this web tool allows the doctor to detect diseases at an early stage.

Keywords—Image Processing, Convolutional Neural Network(CNN), Nail Color Analysis, Disease Detection.

I. INTRODUCTION

In order to find an alternative and easy way in detecting the diseases at the early stages, we go for nail color and texture analysis using Image Processing. Human eye capability has limitations in identifying the colors and also there may be a difficulty because of color blindness whereas computers can identify more than 16 million colors. So, performing nail color analysis through a computer is an effective technology. Pathological tests are complex and painful, while analysis can be performed quickly and still. Since patients are not needed to be present for diagnosis, this approach will benefit patients by allowing doctors to identify problems and issue appropriate prescriptions based only on obtaining nail images.

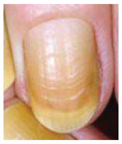
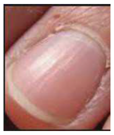
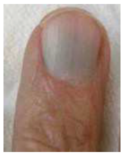
For an image to be defined, it must be composed of rows and columns of pixels. Images may be processed in a variety of ways by translating them into digital form and executing various operations on them. Because human nails have their own unique importance, the proposed system is focused on diagnosing diseases utilizing Finger Nail Color Analysis. The system will detect the various kinds of diseases using RGB values of the nail images by Image Processing Methodology and the tool used is Jupyter Notebook. The acceptance of image can be in any form and finally it gets converted into common format. In the system

- ❖ The picture of a nail is used as input, and an area of interest is manually picked from the image.
- ❖ The color properties of the nail are then extracted by processing the targeted region.
- ❖ Finally, a disease-detection tool is used to match the color characteristic to the algorithm.

As a result, the system aids in the early diagnosis of sickness. This project's main objective is to create a web application that can be used in the healthcare sector because it is affordable and takes less time to analyze..

The description of diseases based on color of the nail image:

IMAGES	NAIL COLORS	POSSIBLE DISEASES
	Pink nails	Normal with good health

	Yellow Nails	Diabetes,Liver Diseases,Digestive problems,Respiratory problems, Vitamin E deficiency
	Pale Nails	Ulcers, malnutrition, kidney, liver, heart, or lung disease, anemia, and other conditions
	Blue Nails	Heart Disease,Hepatitis, Increased Cholesterol,Prior Strokes, Blood clots,Decreased Hemoglobin
	Red Nails	cardiovascular disease, hypertension, pulmonary disease, stroke, and carbon monoxide poisoning

II. RELATED WORKS

For the classification of 11 different nail illnesses, the method in [1] used a variety of deep learning algorithms, including SVM, KNN, and CNN. There are five convolutional and three fully connected layers in our model's architecture, and it achieved a comparable accuracy of 80.45 percent. Using the unique properties of the human palm's nails, the method in [2] is created and executed. Nail color is darker than skin tone, according to a newly discovered algorithm. No machine learning methods are used in this model, which generates an average RGB value from each of the four corners of the palm's backside's four corners, as well as from each of the four top pixels. The accuracy of the model is approximately upto 80%. In [3] the finger nail analysis, in which the color, shape and texture plays a vital role in diagnosing the patients' health status and used grayscale for detecting the disease. The image of the nail is retrieved, and the segmented nail is compared to the healthy nail. If there is any change in color, shape, or texture, the patient is considered unwell. In [4] The input nail image is segmented and analysis is performed on segmented nail area based on the color value. To isolate a specific object in a picture, use image segmentation. Then, utilize detection to assess the segmented region to see if the body is healthy or unhealthy. MATLAB is the software used to create this model. It is possible to determine the kind of illness existing in the human body by analyzing the nail color and texture using MATLAB tools as shown in [5]. We study many methods to improve classification accuracy, including the use of deep learning to detect plant leaf disease, the requirement of big datasets with high variability, data augmentation, transfer learning, and the display of CNN activation maps. We study many methods to improve classification accuracy, including the use of deep learning to detect plant leaf disease, the requirement of big datasets with high variability, data augmentation, transfer learning, and the display of CNN activation maps.[7]. The model divides the conjunctival picture into anemia and non-anemia using the first-order statistical feature extraction method and the K-Nearest Neighbor technique (K-NN).The system has a 71.25 percent level of accuracy. In [8], a disease diagnostic approach involves variations in the color of the nails. A Weka tool is used to extract data from photos of nails for the first training set. When using the nail image's color function, 65% of matching results are accurate. There are preprocessing and segmentation techniques incorporated in [9]. The preparation procedure includes image grayscale conversion, median filtering, edge detection sobel operators, and dilation. K-means region expansion was employed to segment the data. The accuracy of the result is 54.67 %. Using premotor variables (i.e.Olfactory loss and REM sleep behavior disorder (RBD)), a deep learning model was presented in [10] to automatically distinguish between normal persons and Parkinson's Disease patients. A detection accuracy of 96.45% was achieved by the deep learning model provided. A feature significance metric based on the Boosting approach was also produced by the model for Parkinson's Disease identification. Pincer nail patients may be distinguished from healthy persons using a new technique devised by [11] researchers to assist physicians in the early diagnostic process. Using the gravity center fluctuation, the system built a neural network to recognize and explain the characteristics of a pincer nail sufferer. There is a difference between healthy and unhealthy persons in terms of their walking stride, according to the research results. It was developed in [12] that a nail image-based android application might forecast illnesses. To test picture identification and prediction utilizing the most recent and most advanced tensor flow platforms, two different technologies are being employed. On the provided dataset, SVM, KNN, Random Forest, and CNN were all tried in an effort to create a more accurate and efficient prediction system. The comparison analysis demonstrates that CNN, with a testing accuracy of 75% and a training accuracy of 98%, is the best match for the application.

III. PROPOSED WORK

The proposed internet tool helps physicians and patients reduce their burden in an effort to diagnose the condition by analyzing the color of their nails. Because of this, we recommend creating a web application for detecting nail color and reliably predicting illness. The application will predict the various kinds of diseases using RGB values of the nail image by image processing instead of grayscale segmentation. It takes a nail image as an input and will process that nail image and will extract the nail's feature to diagnose the disease. The main objective is to provide a web and mobile application for use in the healthcare domain as it will be cost efficient and consumes less time in analysis.

Convolutional neural networks and no human involvement are used to detect the diseases at an earliest stage using nail color analysis. The job of the user is to take an image of the nail to be screened and that image is being uploaded to the web application in order to detect and predict the disease with a maximum accuracy. The findings are displayed once uploaded photographs are compared to credentials and RGB values that have previously been analyzed using a pre-trained model. Machine learning is used to prepare a dataset for a certain job function.

Supervised and unsupervised machine learning methods are the two main types. The algorithm is trained using labeled inputs and outputs. Machines that learn unsupervised use data that has not been tagged or classified. A type of deep neural network is convolutional neural networks (CNNs). As far as picture recognition goes, CNN is a game changer. They are extensively used in image classification as a back-end tool for analyzing visual information.

NEED FOR THE SYSTEM:

- Computers can classify over 16 million colors with ease; however, the human eye has limitations when it comes to detecting colors, and some people suffer from color blindness. As a result, when compared to the human eye, using a computer to analyze nail color is a superior strategy.
- Pathological tests are difficult and painful, and patients must be available for them, whereas system analysis is simple.

Algorithm of Proposed Model:

Step 1: Input the hand image of the patient.

Step 2: Processing the input nail image.

Step 3: Compare the range values of the color saved in the database to perform Color Analysis.

Step 4: If the color of the nail is red, blue, yellow, pale, white means it will classify as affected nail and corresponding disease is identified and displayed.

Step 5: Otherwise, the output will be a non-affected nail.

Step 6: Finally print the result in the proper format.

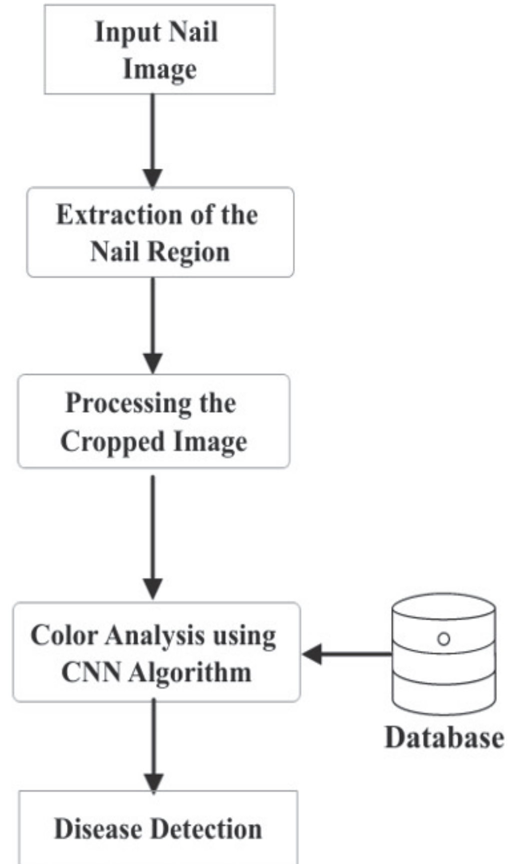


Fig 1: Architecture of Proposed System

1. Database Generation:

In order for machine learning to perform accurately, an image database must be created. The database that comprises a large number of digital photos, such as nail images in PNG format as illustrated in Figures 2 and 3, is manually initialized using a range of information. Because PNG is frequently used for processing and storing full-color images with realistic components and color transitions, that's why we choose it. It has realistic full-color illustrations with numerous color and contrast changes, indicating a high level of picture quality with minimal compression. These attributes collectively explain why the PNG format is so widely used.

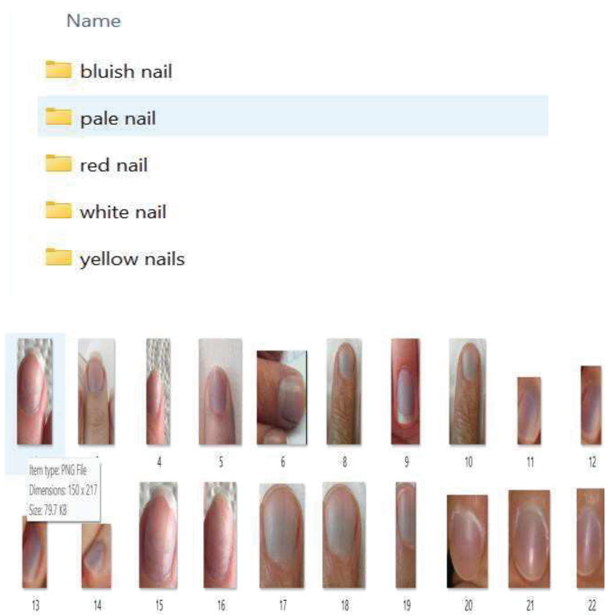
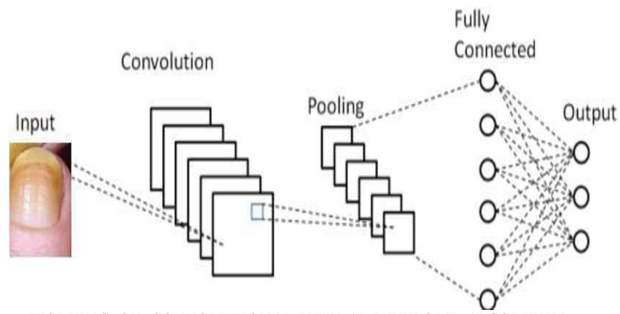
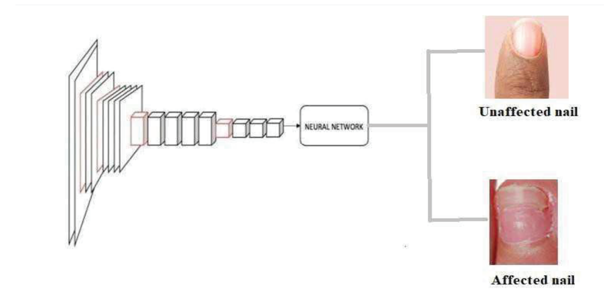


Figure2: Monthly average price of the MSFT share price.



WORKING OF CONVOLUTIONAL NEURAL NETWORK ALGORITHM



NEURAL NETWORK SKELETON WITH LAYERS TO DISTINGUISH AFFECTED NAILS

2.Proposed Algorithm:

DATA INPUT CONSTRAINTS :

- No nail paint or other artificial markings should appear on your fingernails; instead, your nails should be clean and clear.
- Photographs should be shot in well-lit environments.
- There are just two options for the picture background: white or black.
- Images should be shot with no flash and in full daylight.

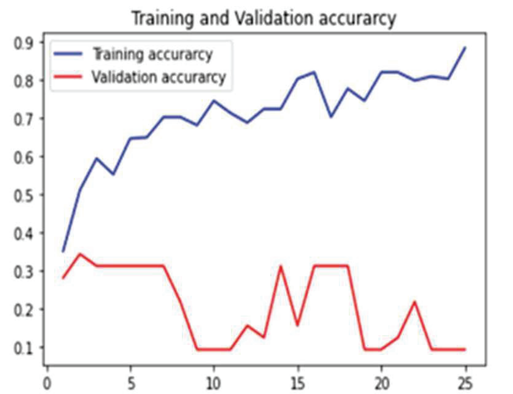
Deep Convolutional Neural Network was used. Images are fed into the Convolutional Neural Network's image classifier, which analyzes the image and categorizes it into many categories. The Convolutional Neural Network contains a number of Hidden layers, which is our initial action on the convolution of an array of pixels of pictures. To overcome the challenge of mismatch, we transferred our output from the previous phase to a layer responsible for reducing the amplitude of pictures. Nail illnesses are then broken down and classified in the second phase. A 19-layer CNN algorithm, trained on thousands of photos in our database, is at the heart of our strategy, which we call "deep learning." According to CNN, CNN has received the earlier output that identifies affected and unaffected nails.

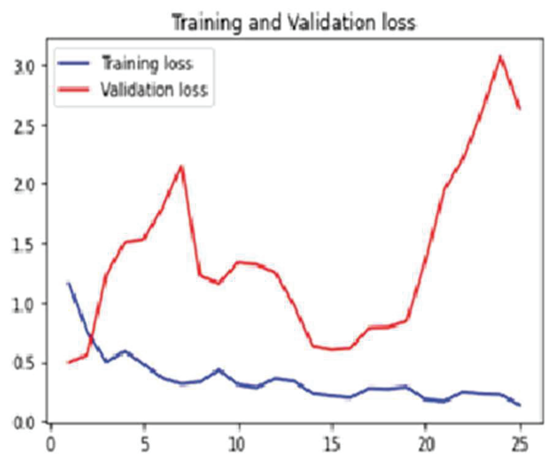
IV. RESULT ANALYSIS AND OUTPUT

```
[INFO] Loading images ...
[INFO] Processing bluish nail ...
[INFO] Processing leukonychia ...
[INFO] Processing pale nail ...
[INFO] Processing red lunula ...
[INFO] Processing yellow nails ...
[INFO] Image loading completed
158
['bluish nail' 'leukonychia' 'pale nail' 'red lunula' 'yellow nails']
[INFO] Spliting data to train, test
Model: "sequential"
_________________________________________________________________
Layer (type)                 Output Shape              Param #
=================================================================
conv2d (Conv2D)              (None, 256, 256, 32)      896
_________________________________________________________________
activation (Activation)      (None, 256, 256, 32)      0
_________________________________________________________________
batch_normalization (BatchNo (None, 256, 256, 32)      128
_________________________________________________________________
max_pooling2d (MaxPooling2D) (None, 85, 85, 32)        0
_________________________________________________________________
dropout (Dropout)            (None, 85, 85, 32)        0
_________________________________________________________________
conv2d_1 (Conv2D)            (None, 85, 85, 64)        18496
_________________________________________________________________
activation_1 (Activation)    (None, 85, 85, 64)        0
_________________________________________________________________
batch_normalization_1 (Batch (None, 85, 85, 64)        256
_________________________________________________________________
conv2d_2 (Conv2D)            (None, 85, 85, 64)        36928
_________________________________________________________________
activation_2 (Activation)    (None, 85, 85, 64)        0
_________________________________________________________________
batch_normalization_2 (Batch (None, 85, 85, 64)        256
_________________________________________________________________
max_pooling2d_1 (MaxPooling2 (None, 42, 42, 64)        0
_________________________________________________________________
dropout_1 (Dropout)          (None, 42, 42, 64)        0
3/3 [==============================] - 6s 2s/step - loss: 0.2788 - acc: 0.7021 - val_loss: 0.7867 - val_acc: 0.3125
Epoch 18/25
3/3 [==============================] - 6s 2s/step - loss: 0.2737 - acc: 0.7766 - val_loss: 0.7922 - val_acc: 0.3125
Epoch 19/25
3/3 [==============================] - 6s 2s/step - loss: 0.2897 - acc: 0.7447 - val_loss: 0.8445 - val_acc: 0.0938
Epoch 20/25
3/3 [==============================] - 6s 2s/step - loss: 0.1840 - acc: 0.8191 - val_loss: 1.3532 - val_acc: 0.0938
Epoch 21/25
3/3 [==============================] - 6s 2s/step - loss: 0.1698 - acc: 0.8191 - val_loss: 1.9497 - val_acc: 0.1250
Epoch 22/25
3/3 [==============================] - 5s 2s/step - loss: 0.2478 - acc: 0.7979 - val_loss: 2.2116 - val_acc: 0.2188
Epoch 23/25
3/3 [==============================] - 5s 2s/step - loss: 0.2322 - acc: 0.8085 - val_loss: 2.6057 - val_acc: 0.0938
Epoch 24/25
3/3 [==============================] - 6s 2s/step - loss: 0.2252 - acc: 0.8021 - val_loss: 3.0717 - val_acc: 0.0938
Epoch 25/25
3/3 [==============================] - 5s 2s/step - loss: 0.1390 - acc: 0.8830 - val_loss: 2.6321 - val_acc: 0.0938
```

```
elif prediction_name == 'clubbing':
        print("Possible diseases are Lung cancer, Lung abscess, Pulmonary tuberculosis, Congestive cardiac failure, Infective enc
elif prediction_name == 'koilonychia':
        print("Possible diseases are iron deficiency anemia")
elif prediction_name == "beau's lines":
        print("Possible diseases are zinc deficiency, pneumonia ")
elif prediction_name == "Muehrck-e's lines":
        print("Possible diseases are arsenic poisoning, thallium poisoning, hodgkins lymphoma, sickle cell, anemia cardiac failu

[[0.00001605 0.00001897 0.9999585  0.00000395 0.0000025 ]]
Predicted Symptome is red nail with an accuracy of: 99.99585
Possible diseases are Systemic lupus erythematosis, Cirrhosis. Alopecia aerate, Psoriasis
```





V. CONCLUSION

By looking at the photographs of the nails, the device can help in the early detection of disorders. A particularly significant skill in the healthcare industry is the capacity to identify various diseases in their early stages. It is simple for doctors to provide patients with the proper treatments thanks to this detecting system. The technology also reduces the amount of time and money needed to detect diseases. This model surpasses the limitations of human vision's resolving capacity, producing findings that are more accurate than human vision.

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